September
09/09/2025
Tuesday

Day - 09
Lecture - 07
Introduction To CSS | Selector

CSS - Cascading Style Sheet

- Changing in font, color, size
- To make website beautiful

Type of CSS -

- Inline
- Internal
- External

① Inline CSS - CSS is written directly inside the HTML tag using <style> attributes

<h1 style ="color:red"> Gym Website </h1>

<h2 style = "background-color: aqua> Exercise </h2>

Weakness : - We have to give style attribute to every element to apply style
- Make HTML code messy and reduce readability.

② Internal CSS - Means you write CSS inside the <style> tag within the <head> section of your HTML file.

```
<head>
    <style>
        h1{
            background-color: aqua;
        }
    </style>
</head>
```

```html
<head>
    <style>
        h2{
            background-color: pink;
        }
    </style>
</head>
```

<u>Note</u>: If website has two same element then it will apply properties on both.

(3) External CSS - Means you write all your CSS in a separate .css file.

- That file is then linked to your HTML file using the <link> tag in the <head>.

HTML file

```html
<html>
<head>
    <title> ... </tite>
    <link rel="stylesheet" href="./style.css">
</head>
<body>
    <h1> Gym Website </h1>
    <h2> Exercise </h2>
</body>
</html>
```

Style.css

```css
h1{
    background-color: pink;
}

h2{
    background-color: aqua;
}
```

Why External CSS is better?

- In inline CSS, if we apply multiply CSS property to one element than code become messy.
- Same for internal CSS, we will write all the properties in the head section.

Both CSS are good for small project

- External CSS
  - We can use the same CSS file across multiple html page.
  - If CSS is written in other file then the HTML code will be cleaner.
  - Faster load time.

* CSS Selectors: Selectors are patterns used to select (target) HTML elements so you can apply styles to then.

(i) Element selector or Tag selector: Selects the HTML elements based on the element name (h1, p, ul)

- Selects all the elements of a specific tag.

Syntax: HTML tag name {

}

Example:

```
h2{
        background-color: red;
}
```

This will select all the <h2> element from HTML file.

(ii) ID Selector: Selects one unique elements with that ID.

- ID should be used only once on a page (Means not 2 element with same ID)

Syntax:

```
#idName{

}
```

<h2 id="first"> Exercise </h2>

Example:

```
#first{
        background-color: blue;
}
```

(iii) Class Selector: It is used when we want to apply same properties to some elements not all.

- Selects elements with specific class name.

Syntax:

```
.ClassName{

}
```

Use dot (.) before Class Name.

```
.third{
        background-color: green;
}
```

(iv) Grouping selector: Select multiple selectors at once using commas.

- Useful when style is same for many elements.

Syntax : element1, element 2, ... {

}

Example : h1, #fourth, .second {

background-color: aqua;

}

* CSS Color : There are many ways through which we can select colors.

(i) Named colors : You write the color by its name.

- Good for basic styling.
- CSS has 140+ predefined colors.

Example- p {

color: red;

}

(ii) RGB color : Stands for (Red Green Blue)

- With the combination of these we can make any color

h1 {

background-color: rgb(159, 19, 161);

We can make custom color

}

- Values goes from (0-255)

RGB (0-255) (0-255) (0-255)

How much memory it will take?

Choose any number between these

127 140 209 → How much bit it will take to store?

↓ ↓ ↓

8 bit 8 bit 8 bit = 24 bits = 3 byte memory will be consume

Why 8-bit?

- Because in binary

255 represented as 1 1 1 1 1 1 1 1 (highest number)

lowest 0 0 0 0 0 0 0 0 (0)

(iii) RGBA - A stands for alpha and represents transparency.

- Value go from 0 to 1 → opaque

0 → transparent

Example -

```
h2 {
    background-color: rgba (127, 149, 64, 0.9);
}
```

(iv) HSL : Stands for Hue, Staturation, lightness

- Hue : The pure color itself. This is a degree on the color wheel (0-360°). 0 - red, 120 - green ~~blue~~, 240 - ~~green~~ blue

- Staturation : It is the intensity of the color. (0% to 100%)

0% → greyscale

100% → fully staturated

• Lightness : Brightness of the color (0% to 100%).

0% → black

50% → normal color

100% → white

Example: h2 {

(0-360) (0-100%)

Color: hsl (80, 80%, 60%);

(0-100%)

}

(v) HSLA : Same A used for transparency

h2 {

Color: hsla (110, 50%, 50%, 0.6);

}

Most used - RGB least used - HSL

(vi) Hexadecimal : Starts with # followed by 6 characters

Example: h2 {

background - color: #639f47;

}

63 9f 47;
Red Green blue

Hexadecimal - We can show 4 bits number

63 9f 47
4 4 4 4 4 4 - 24 bit

Range (0-F) 8 8 8 - 24 bit = 3 byte of memory

0, 1, 2, 3, ... F - We require 4 bits to represent the Hexadecimal number 9 - 1001 F - 1111

* CSS Fonts : Used to style the text

(i) font - size : change the size of the text.

Example :
```
h1 {
        font-size: 20 px;
}
```

- What is the meaning of pixel and how much is this in cm or mm?

pixel full form - picture element

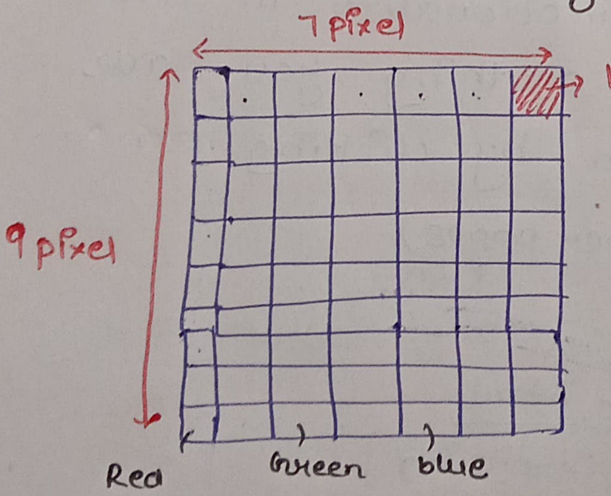
$\boxed{\text{1 pixel is equal to } \frac{1}{96} \text{ inch}}$ This is wrong on google if you search on google

You cannot calculate pixel in cm, m.

What is pixel?

- Smallest unit of image.

- How the image is store in PC?



Known as pixel (very smallest unit is known as pixel)

- Image is divided into rows and columns.

- Every color is stored in the system

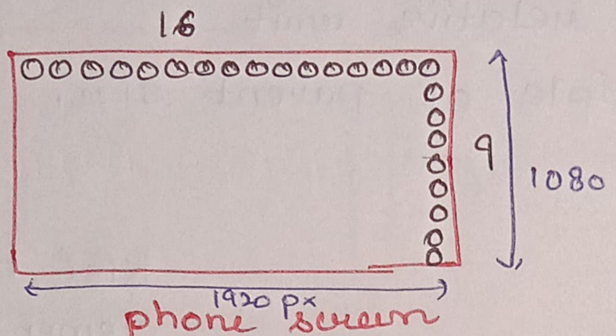
- Every pixel have some color and that small - small boxes store the color and like that the image will be stored.

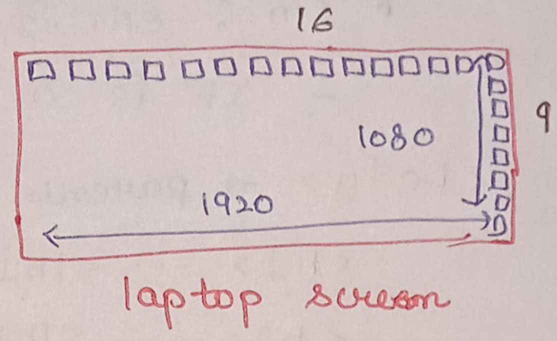
7 * 9 = 63 pixel

1 pixel = RGB (167, 198, 23)

$\underbrace{63}_{\text{Total pixel}} * \underbrace{3 \text{ byte}}_{\text{RGB memory take}} = \underbrace{189 \text{ byte}}_{\text{Total memory}}$

- Pixel size depends on device size





1920 * 1080 pixel x $\underbrace{3\,\text{byte}}_{1\ \text{pixel}}$ 16:9 Resolution

= 6 mb

- In phone 1920 pixel boxes will be small due to phone size
- In laptop boxes (pixel) size will be large because it have to make 1920 pixel in one row.
- So pixel size depends on screen size

So where the concept come 1 pixel = $\frac{1}{96}$ inch

- The person who gave the pixel concept on the basis of his own computer screen size.

And Today we have different screen size.

Don't Do calculation like this and say but my photo size is 1mb in phone

- Some compression techniques are used to store the image size.

Font - size units:

- em : em is the relative unit
  - It is a multiple of parent size.

<body> → parent

<h1> ... </h1>

<h2> --- </h2>

 .. K/li>

parent of → ... → child element of

These are child element

</body>

Kisi bhi element ka parent dhudhna hai toh ek level upar dekho

li {

font - size = $\underbrace{1em}_{\text{parent font-size}}$;

}

font - size = 2em

2 x 20px = 40px

Note: If <ul> is of 20px then <li> will be 20px

At the end this em, rem, % will be converted to pixels

- rem : stands for root em.
  It always depend on root element and root element is <html>

html {

font - size : 16px;

}

h2 {

font - size : 2 rem;

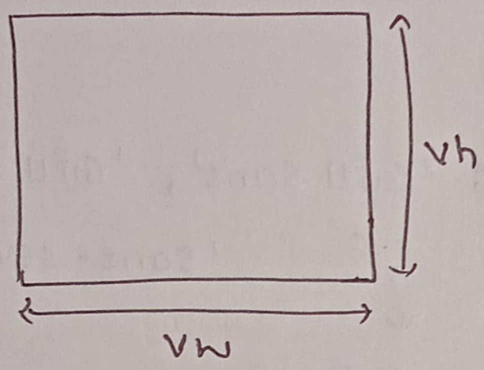
}

↓

2 x 16 = 32px

- vw : Stands for viewport width.
  - It is a relative length unit based on the browser window's width

Viewport : Means visible area on your screen



1vw = 1% of viewport width

h1 {

font-size = 1vw;

}

Suppose your screen is 1000px width.

- 1 vw = 10px
- 50 vw = 500px;
- 100 vw = 1000px (full width)

- vh: viewport height. It is a relative unit based on the browser window's height.

h1 {

font-size : 3 vh;

}

Screen height is 900 px;

1 vh = 9px;

50 vh = 450 px;

- % : A percentage unit defines a size that is relative to the size of the direct parent element.

Parent

ul {

font-size: 50px;

}

li {

Child

font-size: 50%

}

50% means 25 px

(ii) font-family: Set the type of font.

Example: h2 {

font-family: 'Gill Sans', 'Gill Sans MT' 'sans-serif';

}

Sometime this font-doesn't work, then it will apply 2nd. If 2nd not work then 3rd will be apply. This process is known as fallback

(iii) font-weight: Used to control how bold or light text appears.

Example - h1 {

font-weight: 500;

}

Range- (100 to 900)

(iv) font-style: Defines the style of font (normal, italic, etc)

h1 {

font-style: italic;

}

You can also use google font by linking it into your head and then use it in css.